EXPERIMENT 1: DATA PREPROCESSING AND CLEANING

Learning Objectives:

- To clean and preprocess customer churn data.
- To handle missing and duplicate values.
- To encode categorical data.
- To standardize numerical features.
- To visualize relationships in the dataset.

Methods / Python Libraries:

Library / Method	Purpose	Reason for Use
Pandas	Load, clean and analyze data	Easy handling of tabular datasets
NumPy	Numerical operations	Supports numerical calculations
Matplotlib	Create graphs	Creates histograms and plots
Seaborn	Statistical visualizations	Creates count plots, box plots and heatmaps
StandardScaler	Standardize numerical features	Standardizes numerical features

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler

df = pd.read_csv("Customer_Churn.csv")

print(df.head(10))
print(df.tail(10))
print(df.shape)
print(df.columns.tolist())

print(df.info())
print(df.describe(include="all"))

print(df.isnull().sum())
print(df.duplicated().sum())
```

```
df.drop_duplicates(inplace=True)

num_cols = df.select_dtypes(include=["int64","float64"]).columns
cat_cols = df.select_dtypes(include=["object"]).columns

for col in num_cols:
    df[col] = df[col].fillna(df[col].median())

for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])

print(df.isnull().sum())

le = LabelEncoder()

for col in cat_cols:
    df[col] = le.fit_transform(df[col])

scaler = StandardScaler()

df[["Subscription_Length","Monthly_Spend"]] = scaler.fit_transform(
    df[["Subscription_Length","Monthly_Spend"]]
)

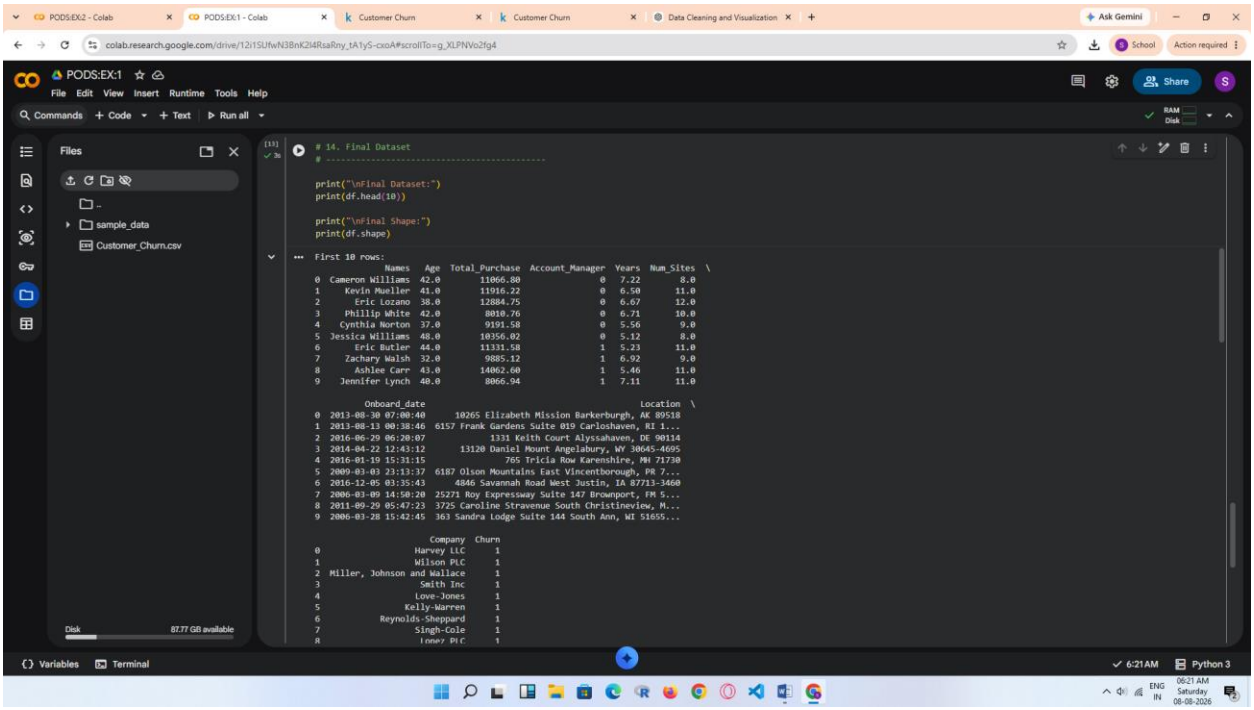
plt.figure(figsize=(8,5))
sns.histplot(df["Monthly_Spend"], kde=True)
plt.title("Distribution of Monthly Spend")
plt.show()

plt.figure(figsize=(8,5))
sns.histplot(df["Subscription_Length"], kde=True)
plt.title("Distribution of Subscription Length")
plt.show()

plt.figure(figsize=(8,5))
sns.countplot(x="Churn", data=df)
plt.title("Churn Count")
plt.show()

plt.figure(figsize=(8,5))
sns.boxplot(x="Churn", y="Monthly_Spend", data=df)
plt.title("Monthly Spend by Churn")
plt.show()
```

Output:



```
# 14. Final Dataset
# ----------------------------------------

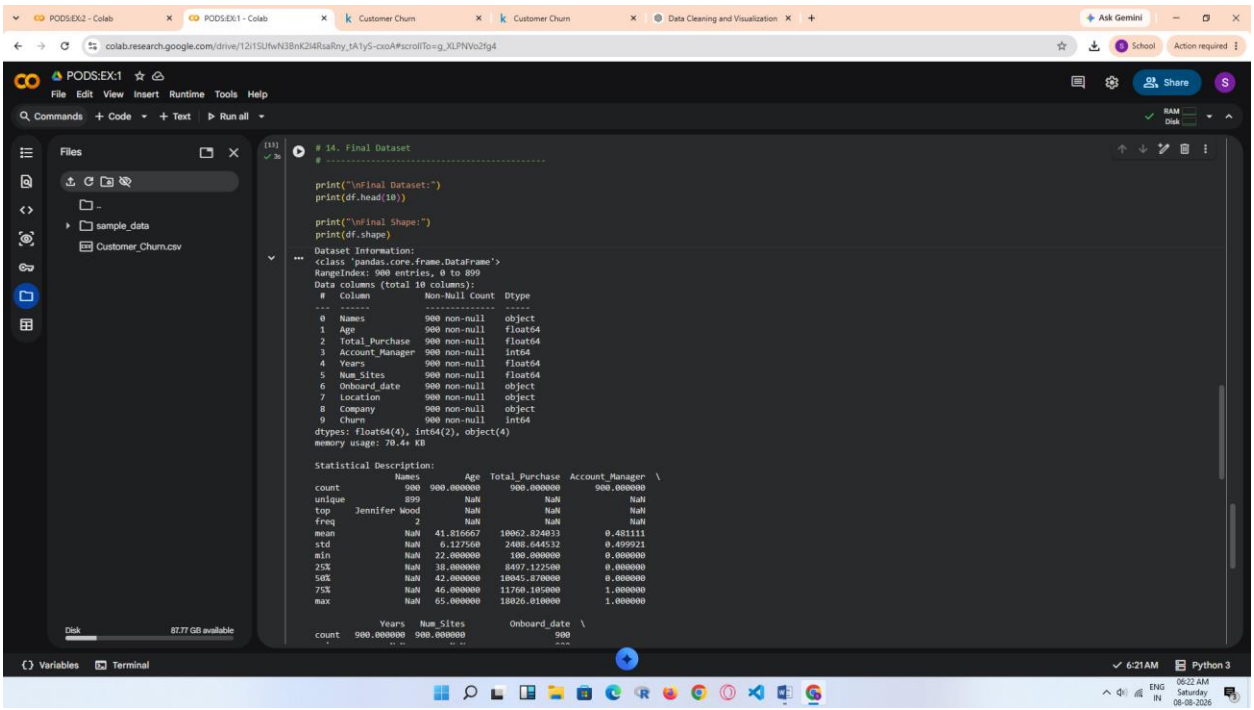
print("\nFinal Dataset:")
print(df.head(10))

print("\nFinal Shape:")
print(df.shape)
```

```
First 10 rows:
              Names   Age  Total_Purchase  Account_Manager  Years  Num_Sites  \
0  Cameron Williams  42.0        11066.80                0   7.22        8.0
1     Kevin Mueller  41.0        11916.22                0   6.50       11.0
2       Eric Lozano  38.0        12884.75                0   6.67       12.0
3    Phillip White  42.0         8010.76                0   6.71       10.0
4    Cynthia Norton  37.0         9191.58                0   5.56        9.0
5  Jessica Williams  48.0        10356.02                0   5.12        8.0
6       Eric Butler  44.0        11331.58                1   5.23       11.0
7    Zachary Walsh  32.0         9885.12                1   6.92        9.0
8      Ashlee Carr  43.0        14062.60                1   5.46       11.0
9    Jennifer Lynch  40.0         8066.94                1   7.11       11.0

         Onboard_date                                           Location  \
0 2013-08-30 07:00:40       10265 Elizabeth Mission Barkerburgh, AK 89518
1 2013-08-13 00:38:46  6157 Frank Gardens Suite 019 Carloshaven, RI 1...
2 2016-06-29 06:20:07             1331 Keith Court Alyssahaven, DE 90114
3 2014-04-22 12:43:12       13120 Daniel Mount Angelabury, WY 30645-4695
4 2016-01-19 15:31:15              765 Tricia Row Karenshire, MH 71730
5 2009-03-03 23:13:17  6187 Olson Mountains East Vincentborough, PR 7...
6 2016-12-05 03:35:43       4846 Savannah Road West Justin, IA 87713-3460
7 2006-03-09 14:50:20  25271 Roy Expressway Suite 147 Brownport, FM 5...
8 2011-09-29 05:47:23  3725 Caroline Stravenue South Christineview, M...
9 2006-03-28 15:42:45  363 Sandra Lodge Suite 144 South Ann, WI 51655...

                        Company  Churn
0                    Harvey LLC      1
1                    Wilson PLC      1
2  Miller, Johnson and Wallace      1
3                     Smith Inc      1
4                    Love-Jones      1
5                   Kelly-Warren      1
6             Reynolds-Sheppard      1
7                    Singh-Cole      1
8                    Toney PLC      1
```



```
# 14. Final Dataset
# ----------------------------------------

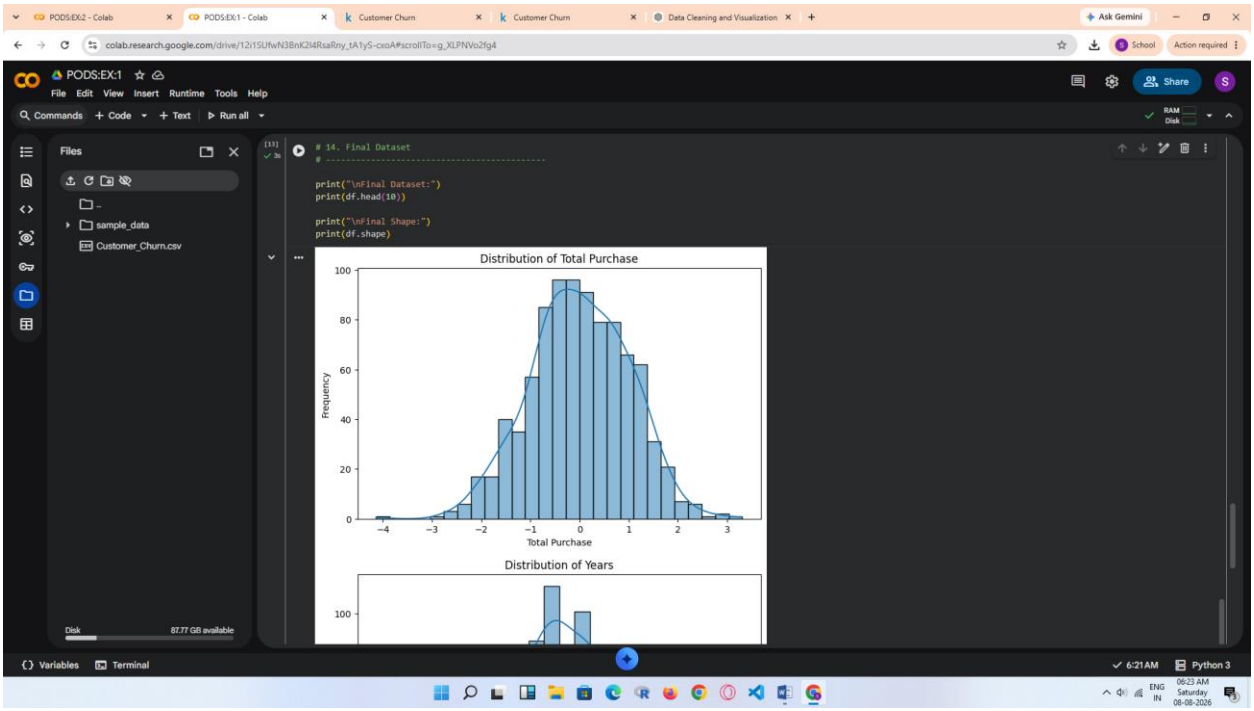
print("\nFinal Dataset:")
print(df.head(10))

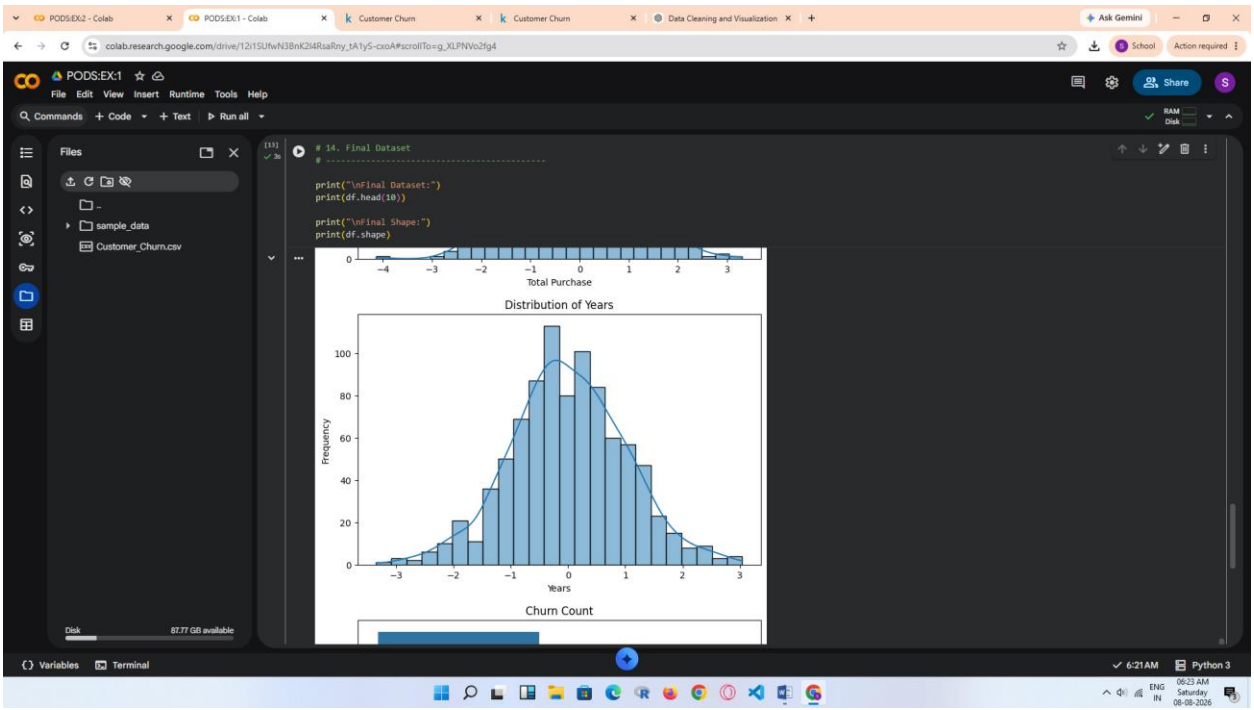
print("\nFinal Shape:")
print(df.shape)
```

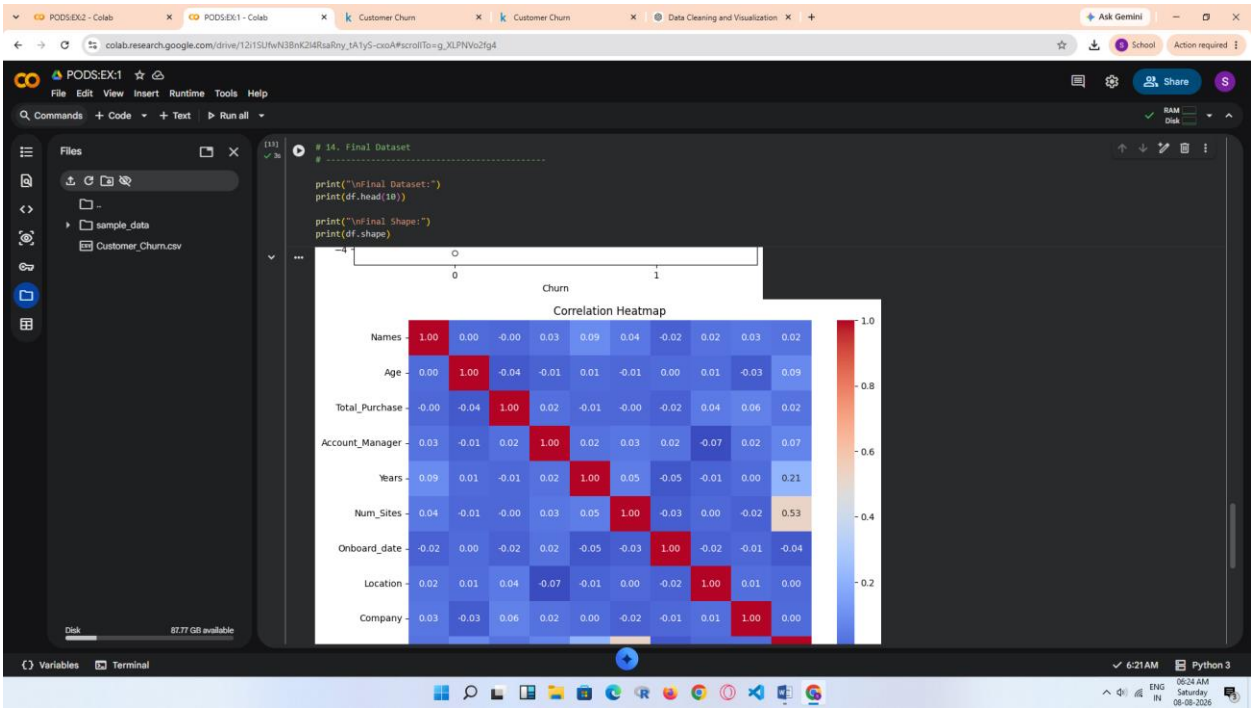
```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 900 entries, 0 to 899
Data columns (total 10 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Names            900 non-null    object
 1   Age              900 non-null    float64
 2   Total_Purchase   900 non-null    float64
 3   Account_Manager  900 non-null    int64
 4   Years            900 non-null    float64
 5   Num_Sites        900 non-null    float64
 6   Onboard_date     900 non-null    object
 7   Location         900 non-null    object
 8   Company          900 non-null    object
 9   Churn            900 non-null    int64
dtypes: float64(4), int64(2), object(4)
memory usage: 70.4+ KB

Statistical Description:
                Names         Age  Total_Purchase  Account_Manager  \
count             900  900.000000      900.000000       900.000000
unique            899         NaN             NaN              NaN
top     Jennifer Wood         NaN             NaN              NaN
freq                2         NaN             NaN              NaN
mean              NaN   41.816667    10062.824033         0.481111
std               NaN    6.127560     2408.644532         0.499921
min               NaN   22.000000      100.000000         0.000000
25%               NaN   38.000000     8497.122500         0.000000
50%               NaN   42.000000    10045.870000         0.000000
75%               NaN   46.000000    11760.105000         1.000000
max               NaN   65.000000    18026.010000         1.000000

             Years   Num_Sites         Onboard_date  \
count   900.000000  900.000000                  900
```


PODS:EX:1
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Files
sample_data
Customer_Churn.csv
Disk 87.77 GB available
14. Final Dataset
print("\nFinal Dataset:")
print(df.head(10))
print("\nFinal Shape:")
print(df.shape)
Distribution of Total Purchase
Frequency
Total Purchase
Distribution of Years
Variables Terminal
6:21 AM Python 3


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Commands + Code + Text Run all
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14. Final Dataset
print("\nFinal Dataset:")
print(df.head(10))
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print(df.shape)
Total Purchase
Distribution of Years
Frequency
Years
Churn Count
Variables Terminal
6:21 AM Python 3



Learning Outcomes:

- Learned to load and explore a customer churn dataset.
- Learned to identify and handle missing values.
- Learned to remove duplicate records.
- Learned to convert data into appropriate datatypes.
- Learned to encode categorical variables.
- Learned to standardize numerical features using `StandardScaler`.
- Learned to visualize data using histograms, count plots, box plots, and heatmaps.